



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 14/11/2019*

PROBLEM

We observe N samples of a realization of the random process $X[n] = A + W[n]$, for $n=0, 1, \dots, N-1$, where A is an unknown deterministic parameter that we want to estimate, and $\{W[n]; n = 0, 1, \dots, N-1\}$ are independent and identically distributed (IID) random variables modeling additive noise. Each noise sample $W[n]$ is modelled by a zero mean Laplace distribution with parameter η as follows:

$$f_w(w) = \frac{1}{2\sigma} e^{-\frac{|w|}{\sigma}}, \quad \sigma > 0, \quad E\{W^2[n]\} = 2\sigma^2.$$

- (1) Derive the estimator of the parameter A by using the Method of moments (MM).
- (2) Derive the bias and the mean square error (MSE) of the MM estimator of A and assess whether it is unbiased and consistent or not.
- (3) Derive the log-likelihood function (LF) and then the maximum likelihood (ML) estimator of the parameter A , assuming that N is an odd integer number. Comment on the computational complexity of the ML estimator with respect to that of the MM estimator.
- (4) Derive the Cramér-Rao lower Bound (CRB) on the estimate of parameter A and assess whether the MM estimator is efficient or asymptotically efficient or not efficient.



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Solution

(1) Derive the estimator of the parameter A by using the Method of moments (MM).

Let's define $X_n \triangleq X[n]$. $\{X_n\}_{n=0}^{N-1}$ independent and identically distributed (IID) random variables, each one with mean A , since the noise samples are zero mean. Hence, the MM estimator is:

$$\hat{A}_{MM} = \frac{1}{N} \sum_{n=0}^{N-1} X[n]$$

which is the well-known sample mean, so it is a linear estimator.

(2) Derive the bias and the mean square error (MSE) of the MM estimator of A and assess whether it is unbiased and consistent or not.

$$E\{\hat{A}_{MM}\} = \frac{1}{N} \sum_{n=0}^{N-1} E\{X[n]\} = \frac{1}{N} \sum_{n=0}^{N-1} A = A, \text{ so the MM estimator is unbiased.}$$

$$MSE\{\hat{A}_{MM}\} = \text{var}\{\hat{A}_{MM}\} = \text{var}\left\{\frac{1}{N} \sum_{n=0}^{N-1} X[n]\right\} = \frac{1}{N^2} \sum_{n=0}^{N-1} \text{var}\{X[n]\} = \frac{1}{N^2} \sum_{n=0}^{N-1} E\{W^2[n]\} = \frac{2\sigma^2}{N}$$

so the MM estimator is consistent.

(3) Derive the log-likelihood function (LF) and then the maximum likelihood (ML) estimator of the parameter A , assuming that N is an odd integer number. Comment on the computational complexity of the ML estimator with respect to that of the MM estimator.

The pdf of the elements of the observed data vector $\mathbf{X}$ is given by:

$$f_X(x_n) = \frac{1}{2\sigma} e^{-\frac{|x_n - A|}{\sigma}}$$

The pdf of the data vector $\mathbf{X}$ in scalar format is:

$$f_{\mathbf{X}}(\mathbf{x}; A) = \prod_{n=0}^{N-1} f_X(x_n; A) = \left(\frac{1}{2\sigma}\right)^N \exp\left(-\frac{1}{\sigma} \sum_{n=0}^{N-1} |x_n - A|\right)$$

$$\ln LF(A) = \ln f_{\mathbf{X}}(\mathbf{x}; A) = -N \ln(2\sigma) - \frac{1}{\sigma} \sum_{n=0}^{N-1} |x_n - A|$$

The ML estimator is derived by solving the likelihood equation.



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$$\frac{d \ln LF(A)}{dA} = -\frac{1}{\sigma} \cdot \frac{d}{dA} \left(\sum_{n=0}^{N-1} |x_n - A| \right) = \frac{1}{\sigma} \sum_{n=0}^{N-1} \text{sign}(x_n - A) = 0$$

$$\sum_{n=0}^{N-1} \text{sign}(x_n - A) = 0$$

The value of A such that the ML equation is satisfied is given by the **sample median** of the data, i.e. we order the data vector $\mathbf{X}$, e.g. with increasing order, and we select the median value. Assuming that N is an odd integer number, the median value is the one at position $(N-1)/2$:

$$\hat{A}_{ML} = \text{median}\{\mathbf{X}\} = Y \left[\frac{N-1}{2} \right],$$

where $Y[n]$, $n = 0, 1, \dots, N-1$ is the ordered vector, such that $Y[0] \leq Y[1] \leq \dots \leq Y[N-1]$.

The ML estimator is nonlinear and requires an ordering of the data vector, so the computational complexity is higher than that of the MM estimator, which is linear.

(4) Derive the Cramér-Rao lower Bound (CRB) on the estimate of parameter A and assess whether the MM estimator is efficient or asymptotically efficient or not efficient.

$$CRB(A) = -\frac{1}{E \left\{ \frac{d^2 \ln LF(A)}{dA^2} \right\}}$$

$$\frac{d^2 \ln LF(A)}{dA^2} = \frac{d}{dA} \left(\frac{d \ln LF(A)}{dA} \right) = \frac{1}{\sigma} \sum_{n=0}^{N-1} \frac{d}{dA} (\text{sign}(x_n - A)) = -\frac{2}{\sigma} \sum_{n=0}^{N-1} \delta(x_n - A)$$

$$\begin{aligned} E \left\{ \frac{d^2 \ln LF(A)}{dA^2} \right\} &= E \left\{ -\frac{2}{\sigma} \sum_{n=0}^{N-1} \delta(x_n - A) \right\} = -\frac{2}{\sigma} \sum_{n=0}^{N-1} \left[\int_{-\infty}^{+\infty} \delta(x_n - A) f_X(x_n) dx_n \right] = -\frac{2}{\sigma} \sum_{n=0}^{N-1} f_X(A) \\ &= -\frac{2N}{\sigma} f_X(A) = -\frac{2N}{\sigma} \cdot \frac{1}{2\sigma} = -\frac{N}{\sigma^2} \end{aligned}$$

$$CRB(A) = -\frac{1}{E \left\{ \frac{d^2 \ln LF(A)}{dA^2} \right\}} = \frac{\sigma^2}{N}$$



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$MSE\{\hat{A}_{MM}\} = \frac{2\sigma^2}{N} = 2 \cdot CRB(A)$, so the MM estimator is not efficient, it is not even asymptotically efficient; whereas we know that the ML estimator is always asymptotically efficient. Hence, asymptotically ($N \gg 1$) we have: $MSE\{\hat{A}_{ML}\} \equiv CRB(A) = \frac{\sigma^2}{N} = \frac{MSE\{\hat{A}_{MM}\}}{2}$.